

THE WASSERSTEIN SEPARATION OF THE BROKEN PHASES: A FINITE-DIMENSIONAL REDUCTION AND THE QUADRATIC CHAOS BOUND

RUQING CHEN

ABSTRACT. For a localized Bost–Connes system in its symmetry-broken phase, the Connes–Rieffel distance $W_1(\varphi_+, \varphi_-)$ between the two extreme KMS_β states was squeezed in Paper IV of the series between $w_0\Pi_\infty(\beta)$ and w_0 , where w_0 is the weight of the group direction and $\Pi_\infty(\beta) = \prod_j \frac{1-t_j}{1+t_j}$, $t_j = \ell_j^{-\beta}$, is the Kakutani product of the order parameter. We sharpen both sides. First, a finite-dimensional reduction: with $W_1^{(N)}$ the distance of the N -coordinate truncation, an exactly computable second-order cone program,

$$\Pi_{(N,\infty)} W_1^{(N)} \leq W_1(\varphi_+, \varphi_-) \leq \Pi_{(N,\infty)} W_1^{(N)} + 2\rho_N,$$

the lower bound exact on truncations and the tail variance $\rho_N^2 = 2 \sum_{j>N} w_j^2 t_j / (1-t_j)^2$ explicit. Second, testing against the degree-one odd Walsh chaos gives the closed-form lower bound

$$W_1(\varphi_+, \varphi_-) \geq \Pi_\infty(\beta) \left(w_0^2 + 4 \sum_j \frac{w_j^2 t_j^2}{(1-t_j)^2} \right)^{1/2},$$

valid whenever $w_0^2 \geq 2 \sum_j w_j^2 t_j / (1-t_j)$ and strictly sharper than $w_0\Pi_\infty(\beta)$ at every finite β . The bound collapses onto the squeeze at the deep-freeze limit $\beta \rightarrow \infty$ and provides an explicit envelope for the second-order vanishing at β_c . We show that near-optimal test functions can be taken asymptotically finite-level and odd under the symmetry, and we record the two questions that remain: a matching upper bound, and whether the optimal profile is parity-measurable.

1. THE SETTING, AND THE SQUEEZE RE-PROVED

Throughout, β lies in the symmetry-broken phase: J is a countable index set of primes ℓ_j with $t_j = \ell_j^{-\beta}$ and $\sum_j t_j < \infty$. The base space is $X = G \times \prod_{j \in J} \bar{\mathbb{N}}$, with G a compact abelian group carrying a fixed character χ of order 2 and a distinguished element $\sigma \in G$ with $\chi(\sigma) = -1$. The symmetric measure is $\nu_{\text{sym}} = \text{Haar} \otimes \bigotimes_j \mu_{t_j}$, where μ_t is the geometric law $\mu_t(k) = (1-t)t^k$ on $\mathbb{N}$. Write $\varepsilon_j(x) = (-1)^{x_j}$, so

$$m_j := \mathbb{E}[\varepsilon_j] = \frac{1-t_j}{1+t_j} \in (0, 1), \quad \Pi_{(N,\infty)} := \prod_{j>N} m_j, \quad \Pi_\infty := \Pi_{(0,\infty)}.$$

Since $\sum_j \mu_{t_j}(x_j \text{ odd}) = \sum_j t_j / (1+t_j) < \infty$, Borel–Cantelli makes all but finitely many ε_j equal to 1 almost surely, so the order parameter

$$\Psi(g, x) = \chi(g) \prod_{j \in J} \varepsilon_j(x_j) \in \{\pm 1\}$$

Date: 5 September 2026.

Paper XXIX of the series on localized Bost–Connes systems. The framework is that of [1], cited by DOI; Paper IV of the series, referred to by numeral, is an unpublished companion, and the one fact used from it (the squeeze theorem) is re-proved in place.

is defined ν_{sym} -a.e., and $\nu_{\text{sym}}(\Psi = \pm 1) = \frac{1}{2}$ by the χ -symmetry of Haar measure. The extreme states are $\varphi_{\pm}(f) = 2 \int f \mathbf{1}_{\{\Psi=\pm 1\}} d\nu_{\text{sym}}$, so that

$$(1) \quad \varphi_+(f) - \varphi_-(f) = 2 \mathbb{E}[f \Psi].$$

The Lipschitz seminorm, in the Rieffel formalism [2], is

$$L(f) = \left\| \left(w_0^{-2} |\Delta_0 f|^2 + \sum_{j \in J} w_j^{-2} |\Delta_j f|^2 \right)^{1/2} \right\|_{\infty},$$

with $\Delta_0 f(g, x) = f(g\sigma, x) - f(g, x)$ the group difference, $\Delta_j f$ the one-step difference in the j -th coordinate, and weights satisfying $\sum_j w_j^2 < \infty$. The distance is $W_1(\varphi_+, \varphi_-) = \sup\{\varphi_+(f) - \varphi_-(f) : f \in C(X)_{sa}, L(f) \leq 1\}$.

Lemma 1.1 (Squeeze, re-proved). $w_0 \Pi_{\infty}(\beta) \leq W_1(\varphi_+, \varphi_-) \leq w_0$.

Proof. Upper: the map $T(g, x) = (g\sigma, x)$ preserves ν_{sym} and flips Ψ , hence transports $\nu_+ := 2\mathbf{1}_{\{\Psi=1\}}\nu_{\text{sym}}$ to ν_- ; for $L(f) \leq 1$ one has $|f \circ T - f| = |\Delta_0 f| \leq w_0$ pointwise, so $|\varphi_+(f) - \varphi_-(f)| = |\int (f - f \circ T) d\nu_+| \leq w_0$. Lower: $f = \frac{w_0}{2}\chi$ has $\Delta_0 f = -w_0\chi$, all other differences zero, so $L(f) = 1$, and (1) gives $2\mathbb{E}[f\Psi] = w_0 \mathbb{E}[\chi^2 \prod_j \varepsilon_j] = w_0 \Pi_{\infty}$. $\square$

2. THE FINITE-DIMENSIONAL REDUCTION

Fix N and let $\mathcal{F}_{\leq N}$ be the σ -algebra generated by g and $x_1, \dots, x_N$, with E_N the corresponding conditional expectation (integrating out $x_j, j > N$). Write $\Psi_N = \chi \prod_{j \leq N} \varepsilon_j$ and $T_N = \prod_{j > N} \varepsilon_j$, so $\Psi = \Psi_N T_N$ with T_N independent of $\mathcal{F}_{\leq N}$ and $\mathbb{E}[T_N] = \Pi_{(N, \infty)}$. Define the truncated distance

$$W_1^{(N)} := \sup\{2 \mathbb{E}[f \Psi_N] : f \text{ } \mathcal{F}_{\leq N}\text{-measurable, } L(f) \leq 1\}.$$

Theorem 2.1 (Sandwich by truncations). *For every N ,*

$$\Pi_{(N, \infty)} W_1^{(N)} \leq W_1(\varphi_+, \varphi_-) \leq \Pi_{(N, \infty)} W_1^{(N)} + 2\rho_N, \quad \text{where } \rho_N^2 := 2 \sum_{j > N} \frac{w_j^2 t_j}{(1 - t_j)^2} \rightarrow 0.$$

The lower bound is exact on truncations: for an optimizer f_N of $W_1^{(N)}$ the value $2 \mathbb{E}[f_N \Psi]$ equals $\Pi_{(N, \infty)} W_1^{(N)}$ exactly.

Proof. Lower. For $\mathcal{F}_{\leq N}$ -measurable f , independence gives $\mathbb{E}[f\Psi] = \mathbb{E}[f\Psi_N] \mathbb{E}[T_N]$, which is $\Pi_{(N, \infty)} \mathbb{E}[f\Psi_N]$; take the supremum.

Upper. Split $f = E_N f + (f - E_N f)$. For $j \leq N$ the difference Δ_j commutes with E_N , and $\Delta_j E_N f = 0$ for $j > N$; by the vector-valued Jensen inequality applied to the ℓ^2 -norm inside the supremum, $L(E_N f) \leq L(f) \leq 1$, so the first term contributes at most $\Pi_{(N, \infty)} W_1^{(N)}$ by the computation above. For the second term, expand $f - E_N f = \sum_{j > N} D_j f$ into martingale differences [3, Ch. 3] along the coordinates $j > N$ (in any fixed order); the increments are orthogonal, and $|D_j f| \leq \mathbb{E}_{x'_j} |f(\dots, x_j, \dots) - f(\dots, x'_j, \dots)| \leq w_j \mathbb{E}|x_j - x'_j|$ pointwise, since $L(f) \leq 1$ forces $|\Delta_j f| \leq w_j$ at every point and the coordinate can be moved one step at a time. Hence

$$\mathbb{E}|f - E_N f|^2 = \sum_{j > N} \mathbb{E}|D_j f|^2 \leq \sum_{j > N} w_j^2 \mathbb{E}[(x_j - x'_j)^2] = \sum_{j > N} w_j^2 \cdot 2 \text{Var}(x_j) = \rho_N^2,$$

using $\text{Var}_{\mu_t}(x) = t/(1-t)^2$. Since $|\Psi| = 1$, Cauchy-Schwarz bounds the second term's contribution to (1) by $2\rho_N$. $\square$

Remark 2.2 (The truncation is a second-order cone program). $W_1^{(N)}$ maximizes the linear functional $2\mathbb{E}[f\Psi_N]$ of the finitely many values of f subject to one second-order cone constraint per point of the (truncated) configuration space: after additionally truncating each coordinate to $x_j \leq M$ — an error controlled by the geometric tails $\sum_{j \leq N} w_j t_j^M / (1 - t_j)$ through the same martingale argument — this is a standard SOCP with $O(2 \cdot M^N)$ variables (the group coordinate enters only through the two values of χ), solvable exactly by interior-point methods. Theorem 2.1 converts any numerical solution into a certified two-sided bound on W_1 .

3. THE QUADRATIC CHAOS BOUND

We now evaluate the degree-one odd Walsh chaos exactly.

Theorem 3.1 (Quadratic sum lower bound). *Set $\kappa_j := \frac{1 - m_j}{m_j} = \frac{2t_j}{1 - t_j}$, and suppose*

$$(2) \quad w_0^2 \geq \sum_{j \in J} w_j^2 \kappa_j.$$

Then

$$W_1(\varphi_+, \varphi_-) \geq \Pi_\infty(\beta) \left(w_0^2 + \sum_{j \in J} w_j^2 \kappa_j^2 \right)^{1/2} = \Pi_\infty(\beta) \left(w_0^2 + 4 \sum_{j \in J} \frac{w_j^2 t_j^2}{(1 - t_j)^2} \right)^{1/2},$$

the series converging by $\sum_j w_j^2 < \infty$ and $t_j \rightarrow 0$. The bound is strictly larger than the squeeze lower bound $w_0 \Pi_\infty(\beta)$ at every $\beta < \infty$, and collapses onto it as $\beta \rightarrow \infty$.

Proof. Take the ansatz $f(g, x) = \chi(g)(c + \sum_j r_j \varepsilon_j(x_j))$ with $c, r_j \geq 0$ and $\sum_j r_j < \infty$. Its differences are $\Delta_0 f = -2\chi(c + \sum_j r_j \varepsilon_j)$ and $\Delta_j f = \chi r_j \Delta_j \varepsilon_j$ with $|\Delta_j \varepsilon_j| = 2$, so the Lipschitz constraint reads, pointwise in the signs $\varepsilon \in \{\pm 1\}^J$,

$$\frac{4(c + \sum_j r_j \varepsilon_j)^2}{w_0^2} + \sum_j \frac{4r_j^2}{w_j^2} \leq 1.$$

The supremum over sign patterns is attained at $\varepsilon_j \equiv +1$, so with $s := c + \sum_j r_j$ the constraint is exactly the ellipsoid $s^2/(w_0/2)^2 + \sum_j r_j^2/(w_j/2)^2 \leq 1$. Independence and $\varepsilon_j^2 = 1$ give the objective

$$2\mathbb{E}[f\Psi] = 2 \left(c \Pi_\infty + \sum_j r_j \frac{\Pi_\infty}{m_j} \right) = 2\Pi_\infty \left[s + \sum_j r_j \kappa_j \right],$$

after substituting $c = s - \sum_j r_j$. Maximizing the linear functional $s + \sum_j r_j \kappa_j$ over the ellipsoid gives the value $\frac{1}{2}(w_0^2 + \sum_j w_j^2 \kappa_j^2)^{1/2}$, attained at $s^* = \frac{(w_0/2)^2}{R}$, $r_j^* = \frac{(w_j/2)^2 \kappa_j}{R}$ with $R = \frac{1}{2}(w_0^2 + \sum_j w_j^2 \kappa_j^2)^{1/2}$; the sign conditions $r_j^* \geq 0$ hold, and $c^* = s^* - \sum_j r_j^* \geq 0$ is exactly (2). Multiplying by $2\Pi_\infty$ yields the bound. Strictness over $w_0 \Pi_\infty$ holds whenever some $t_j > 0$, i.e. at every finite β ; as $\beta \rightarrow \infty$ every $\kappa_j \rightarrow 0$ and the correction vanishes. $\square$

Remark 3.2 (The boundary case). If (2) fails — possible near β_c when the weights decay slowly — the maximum of the same linear functional is taken on the face $c = 0$ of the cone, i.e. on the ellipsoid intersected with $\{\sum_j r_j = s\}$; this is again a closed-form Lagrange computation, and the resulting bound still dominates $w_0 \Pi_\infty(\beta)$, but we do not display the longer formula. In either regime the theorem provides an explicit lower envelope for the second-order vanishing of W_1 at β_c established in Paper IV.

4. STRUCTURE OF OPTIMIZERS, AND WHAT REMAINS

Proposition 4.1 (Odd, asymptotically finite-level). *The supremum defining W_1 is unchanged if restricted to functions that are odd under $T(g, x) = (g\sigma, x)$, and for every $\epsilon > 0$ it is attained within ϵ by an $\mathcal{F}_{\leq N}$ -measurable function as soon as $2\rho_N \leq \epsilon$.*

Proof. The odd part $f^\circ = \frac{1}{2}(f - f \circ T)$ satisfies $\mathbb{E}[f^\circ \Psi] = \mathbb{E}[f \Psi]$ (since $\Psi \circ T = -\Psi$ and ν_{sym} is T -invariant) and $L(f^\circ) \leq L(f)$ (each difference of f° is the average of two differences of f). The second statement is Theorem 2.1 with $E_N f$ in place of f . $\square$

Conjecture 4.2. The increasing sequence of values of the odd Walsh-chaos programs — degree one being Theorem 3.1, degree d allowing coefficients $r_A \prod_{j \in A} \varepsilon_j$ with $|A| \leq d$ — converges to $W_1(\varphi_+, \varphi_-)$.

Remark 4.3 (Numerical verdict). The programs of Remark 2.2 were solved for a toy model ($w_0 = 1$, $w_j = (j+1)^{-1}$, $t_j = p_j^{-1}$, $N \leq 3$; the feasible set is convex and the objective linear, so any KKT point is the global maximum, and constraint violations were at machine precision). Two findings, scripts and outputs in `code/conj42/` of the repository. First, the degree-one bound of Theorem 3.1 is far from sharp: the exact 2^N -point parity program — whose value is the limit of the odd Walsh-chaos hierarchy, since a forward step always flips parity and the constraint closes on parity classes without any truncation — exceeds it by 26% at $N = 2$ and 39% at $N = 3$. Higher chaos participates massively. Second, and against Conjecture 4.2: the full grid optimum exceeds the parity optimum, in matched truncations, by a stable $2\text{--}4 \times 10^{-3}$ that is independent of the grid depth M at $N \geq 2$, while at $N = 1$ (where parity-measurable and degree-one coincide) the same gap vanishes to 10^{-15} once the boundary parity is matched — a control showing the $N \geq 2$ gap is a bulk effect, not a boundary artifact. The evidence is therefore that the optimal profile is *not* parity-measurable and that the odd-chaos hierarchy converges to a value strictly below $W_1^{(N)}$: Conjecture 4.2 is expected to be false as stated, and the parity value should be read as a computable, certified lower bound rather than the answer.

Two questions remain, and they are the genuine content left in this problem. First, a matching *upper* bound: the transport map of Lemma 1.1 moves mass only in the group direction; any improvement must couple the group move to the parity configuration, an x -adaptive coupling whose cost functional is not yet controlled. Second, *parity measurability* (on which Remark 4.3 already gives a numerical answer): the order parameter sees only the parities ε_j , but conditioning a test function on the parity σ -algebra does not obviously contract L , because the one-step differences do not respect the parity fibres; whether genuinely non-parity-measurable test functions can strictly increase the supremum is open. A numerical test of Conjecture 4.2 is available today: Remark 2.2 certifies two-sided bounds for small N , to be compared with the closed form of Theorem 3.1.

REFERENCES

- [1] R. Chen, *KMS state uniqueness and phase transitions in localized Bost–Connes systems*, preprint, 2026, <https://doi.org/10.5281/zenodo.22152101>.
- [2] M. A. Rieffel, *Metrics on state spaces*, Doc. Math. 4 (1999), 559–600.
- [3] S. Boucheron, G. Lugosi, P. Massart, *Concentration Inequalities: A Nonasymptotic Theory of Independence*, Oxford University Press, 2013.

GUT GEOSERVICE INC., MONTREAL, CANADA
 Email address: ruqing@hotmail.com